ABSTRACT

Decentralized Finance (DeFi), an ecosystem of blockchain-based financial applications that operate without traditional intermediaries**,** has developed a wide range of mechanisms through which digital assets generate yield. For most of its history, however, that yield could not be traded on its own: it was simply the outcome of holding a position. This dissertation studies yield tokenization, the separation of a yield-bearing asset into a claim on the principal and a claim on the future yield, each tradable independently. Its central question is whether yield tokenization represents a step in the maturation of DeFi toward a possible crypto-native fixed income market. DeFi grew out of a technology-driven environment, and yield tokenization is examined as a point at which it begins to adopt the concepts and conventions of traditional fixed income. The Pendle protocol, the leading application of this idea, is examined as a case study through a qualitative and conceptual analysis based on the academic literature, the protocol's whitepapers and documentation, and its public code. The analysis develops an imperfect analogy between these tokenized instruments and traditional fixed income, including instruments bought at a discount and redeemed at maturity, rates of return implied by market prices, and maturities that together form something comparable to a yield curve. It also identifies the points at which the analogy stops holding, since the two sets of instruments exist in different environments, one consolidated by institutions over a long history, the other young, built in code, and still defining its rules. It then shows that supporting such instruments required more than creating the tokens: the protocol had to redesign the automated market maker through which they trade, so that time to maturity and rates of return, rather than only prices, are built into the trading mechanism. This adaptation brings on-chain trading closer to the conventions under which fixed income is traded. It further examines the risks specific to tokenized yield, including smart contract, liquidity, collateral, governance and regulatory risks. The findings suggest that yield tokenization should be understood not merely as a new class of financial instruments, but as an early stage in the formation of a crypto-native fixed income market structure. On this basis, the dissertation argues that Pendle is best understood not as a single instrument but as a market structure assembled inside a protocol, encompassing issuance, pricing, trading and the alignment of the tokens with the yield they represent. Within the protocol, this structure is organized in four layers: the instrument, pricing by yield, liquidity infrastructure, and the convergence and arbitrage mechanisms that keep rates aligned.

Together, they seek to reproduce on-chain the functions that a fixed income market requires, from issuance to price formation. A further contribution lies in the approach itself. Digital finance is a hybrid field, neither purely financial nor purely technological. It resists the familiar categories through which each side tries to read it, and most of the literature still examines it from one side only. By integrating concepts from traditional finance, blockchain infrastructure and DeFi protocol design into a single analytical framework, the work treats the two perspectives as inseparable.

Keywords: Decentralized Finance (DeFi), Yield Tokenization, Pendle Protocol, Fixed Income, PT/YT, On-Chain Yield, Crypto Markets, DeFi Interest Rates, Smart Contract Risk.